

Raaghav Thirumaligai

Mechanical Engineer · Controls Researcher · Prototype Designer

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PROFILE

Mechanical engineer working across control theory, physical prototyping, and product design. Builds models from physical systems and experimental data, develops control and estimation algorithms, and turns concepts into functional hardware and software prototypes. Experience spans electric-vehicle systems, robotics, research instrumentation, interaction concepts, and optimal control. Combines an M.S. in thermofluids with ongoing Ph.D. in control theory and hands-on design, fabrication, and testing.

EDUCATION

University of California, Santa Barbara – Ph.D. Mechanical Engineering **Projected Jun 2029**
In progress; first year completed. Control Theory. GPA: 3.92/4.0.
Open to taking academic leave for the right full-time opportunity.

University of California, Santa Barbara – M.S. Mechanical Engineering **Jul 2025**
Thermofluids. GPA: 3.90/4.0. Project: clogging of fiber suspensions at constrictions.

University of California, Santa Barbara – B.S. Mechanical Engineering **Jun 2025**
Product Design. GPA: 3.76/4.0. Capstone: [600 V Vehicle Battery Container](#).

TECHNICAL SKILLS & METHODS

Controls & Optimization	State-space systems, LQR/LQG, MPC and economic MPC, constrained and trajectory optimization, nonlinear control, Lyapunov stability, robust control, H^∞ synthesis, hybrid systems
Modeling & Estimation	System identification, observers, Kalman filtering, distributed-parameter models, finite-difference discretization, balanced reduction and truncation, stability and sensitivity analysis
Mechanical Design	SolidWorks, Fusion 360, assemblies, engineering drawings, GD&T, FEA, sheet metal, fixtures, laser/waterjet fabrication, FDM/SLA 3D printing
Prototyping & Interaction	Physical mockups, hand sketches, CAD renders, Keynote storyboards and animation, haptic/touch concepts, sensor-driven input/output behavior
Programming & Data	MATLAB/Simulink, Python, OpenCV, Swift/SwiftUI, Git; C++20 (basic); classifier training, vision-based detection, experimental data analysis
Hardware & Experiments	Simulink/QUARC, Arduino, Raspberry Pi, sensors and actuators, DAQ, strain gauges, calibration, filtering, experimental test design, state-machine logic

ENGINEERING, ROBOTICS & LEADERSHIP

Gaucha Racing Formula SAE EV – Brake Systems Lead / Vice President **Aug 2021 – Jun 2025**

- Four-year contributor on an electric race-car design-build-test team; led work across vehicle systems and coordinated teammates through build cycles.
- Designed and built a 600 V accumulator container using an adhesive-first joining strategy; produced CAD, engineering drawings, and laser-cut sheet metal manufacturing.
- Co-led brake-system design, manufacturing, testing, and reliability work; separately led a funded study of brake-fluid viscosity and presented the results.

Autonomous Can-Collection Robot (C.L.E.A.N.) – Team Lead **Apr 2024 – Jun 2024**

- Collaborated with a five-person team through weekly design-build-test cycles to deliver a robot that found, approached, identified, grasped, and lifted aluminum cans.
- Trained and tuned an OpenCV cascade classifier on approximately 1,500 custom images; converted the detected can's bounding-box centroid into Raspberry Pi steering commands.
- Integrated a custom 3D-printed treaded chassis and 2-DOF worm-gear arm/claw with motor drivers, an Arduino, and ultrasonic and inductive sensors.

CONTROL SYSTEMS, MODELING & OPTIMIZATION PROJECTS

Reduced-Order LQG Control of a Distributed-Parameter System

Apr 2026 – Jun 2026

- Built a 99-state finite-difference model of a boundary-actuated damped-wave PDE in MATLAB; augmented it with frequency-shaped disturbances and measurement noise for LQG output-feedback design.
- Designed LQR/Kalman controllers for three disturbance bandwidths, then used balanced truncation to test reduced-order controllers (orders 1–10) on the full plant across damping regimes.

Constrained Satellite Rendezvous with MPC

Jan 2026 – Mar 2026

- Built a Hill–Clohessy–Wiltshire relative-motion model and derived a zero-order-hold discrete-time state-space system for receding-horizon control in MATLAB/MPCTools.
- Implemented and benchmarked saturated LQR, state-focused constrained MPC, and economic MPC with different state, actuator, and fuel priorities.
- Evaluated convergence, feasibility, and a fuel proxy over randomized initial conditions; presented the resulting speed-versus-energy tradeoffs in a final report and slides.

Inverted Pendulum on Cart – System ID, Estimation & Hybrid Control

Apr 2025 – Jun 2025

- Collected logarithmic sine-sweep data on a physical system; identified linear models around stable and unstable equilibria and checked them with frequency-response and root-locus analysis.
- Implemented energy-based swing-up, LQR/LQG balance and catch, Kalman-filter estimation, and state-machine mode switching in MATLAB/Simulink with QUARC real-time hardware execution.

King–Rook–King Endgame – Noncooperative Game Theory

Feb 2026 – Mar 2026

- Modeled the chess endgame as a finite, sequential, zero-sum dynamic game in MATLAB.
- Implemented legal-move generation, checkmate/stalemate detection, and depth-limited minimax.
- Designed cost function using terminal rewards/penalties and black-king mobility heuristics.

PRODUCT CONCEPTS & INTERACTIVE SOFTWARE

Rapid Interaction Studies – Disc, Orb & Hover Touch

Aug 2026

- Developed three speculative hardware-interaction concepts through hand sketches, Keynote storyboards, animation, and a rapid physical mockup.
- Built a same-day cardboard Disc prototype to test palm scale, grip, press location, thickness, and display glare before committing to CAD or code.

Frame – Ambient iPadOS Dashboard (SwiftUI)

Aug 2026 – Present

- Carried a spare-iPad concept from sketch and interaction architecture into a functional native application and public case study, with a glanceable ambient layer, progressive disclosure, and a camera-first feed carousel.
- Designed thoughtful and hopefully intuitive ambient interface, revealing relevant information, integrating sources from multiple Apple Frameworks.

RESEARCH & EXPERIMENTAL SYSTEMS

Biological Control Lab, Yeung UCSB – Research Assistant

Jan 2025 – Jan 2026

- Co-developed a control-oriented model of genome-wide transcription coupled to DNA-supercoiling dynamics regulated by topoisomerases.
- Supported stability and sensitivity analysis of stochastic and nonlinear population models to identify parameters influencing transcription and growth dynamics.
- Co-author on a manuscript accepted in January 2026; [preprint: doi:10.1101/2025.05.26.655193](#).

M&M Flow Lab, Sauret UCSB – Master’s Researcher

Aug 2024 – Jun 2025

- Studied clogging dynamics in particle-laden suspensions at constrictions, where stochastic events, changing material conditions, and limited observability complicated measurement.
- Prototyped a Python vision detector using a rolling brightness/windowed-visibility metric to identify occlusion events and support experiment logging and post-analysis.
- Presented [clogging of fiber suspensions at constrictions](#) at the November 2024 APS Division of Fluid Dynamics meeting.

M&M Flow Lab, Sauret UCSB – Undergraduate Researcher

Jan 2023 – Aug 2024

- Explored cohesive forces in granular materials with two Ph.D. student mentors to study inter-particle mechanics.
- Designed and built an instrumented annular shear cell, integrating mechanical hardware and strain-gauge measurements.
- Developed an Arduino–MATLAB DAQ, calibration, and filtering workflow for repeatable experiments; presented results at the Southern California Flow Physics Symposium.

Fluid Energy Science Lab, Luzzatto-Fegiz UCSB – Undergraduate Researcher

Jun 2023 – Oct 2023

- Collected wind-tunnel data for wake-model verification and built an Arduino–MATLAB DAQ system and app.
- Fabricated laser-cut and 3D-printed annuli for repeatable wind-energy experiments.

TEACHING & ADDITIONAL EXPERIENCE

UCSB Mechanical Engineering – Teaching Assistant

Apr 2024 – Present

- Led mechanical-engineering laboratory sections covering experimental procedures and data analysis; taught dynamics through office hours on free-body diagrams and energy-based system analysis.
- Supported vibrations and fluids instruction through grading, office hours, and technical coaching.
- Advised bio-inspired design groups and communicated unfamiliar technical ideas across different learning styles.

UCSB San Rafael Residence Hall – Resident Assistant

Aug 2022 – Jun 2023

- Managed resident support, conflict resolution, and event planning in a high-traffic residence hall.

Diablo Escapes, Walnut Creek – GameMaster

Jul 2020 – Aug 2022

- Guided customers through VR escape rooms, troubleshooted hardware/software issues, and explained unfamiliar technology.
- Helped design puzzles for physical escape rooms under construction.

HONORS & RESEARCH AWARDS

- Undergraduate Engineering Honors Program, September 2023 – June 2024.
- Dean’s Honors (Engineering), five quarters.
- Two Undergraduate Research Awards, \$3,750 and \$2,500 (2024 and 2025).